

MECANISMO-DE-FRENO-DE-TAMBOR.pdf



laktecuento



Cinemática y Dinámica de Máquinas



3º Grado en Ingeniería Mecánica



Escuela de Ingeniería de Barcelona este (Eebe)
Universidad Politécnica de Catalunya



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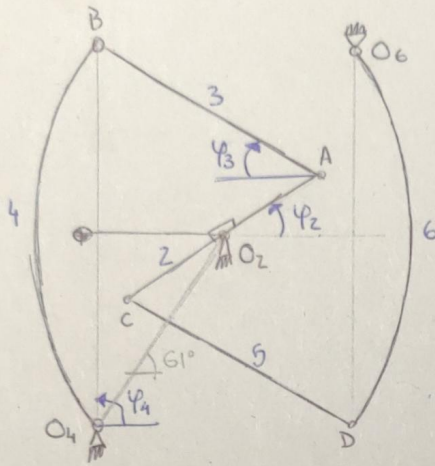
El momento villancico
en familia, no renta.

webuy.com

CAMBIAR
tu Play 4 por una
Play 5 y jugar
con amigos, sí.



PROBLEMA 3.3. MECANISMO DE FRENO DE TAMBOR.



$\varphi_2 = 29^\circ \rightarrow$ posición en reposo

$\varphi_2 = 50^\circ \rightarrow$ freno se activa.

Acceleración del CDG
del eslabón 2 si:

$$\ddot{\varphi}_2 = 15 \text{ rad/s}^2$$

$$\dot{\varphi}_2 = 10 \text{ rad/s}$$

$$\vec{q} = [\varphi_2, \varphi_3, \varphi_4]^T$$

DATOS GEOMÉTRICOS:

$$O_2A = 49 \text{ mm}$$

$$AC = 98 \text{ mm}$$

$$AB = 100 \text{ mm}$$

$$O_4B = 153 \text{ mm}$$

$$O_4O_2 = 87 \text{ mm}, 61^\circ$$

EL ESABÓN 4 ES UN SECTOR
CIRCULAR DE RADIO 90 mm Y ÁNGULO
110° CON CENTRO A +14 mm de O_2

$$H = 42.2 \text{ mm} \rightarrow 87 \cos(61)$$

$$V = 76.1 \text{ mm} \rightarrow 87 \sin(61)$$

* POSICIÓN

$$m_g = 3 - 1 = 2 \text{ E.C.S.}$$

* E.E.G.'S

$$\Phi_q = \begin{cases} O_4B \cdot \cos(\varphi_4) + AB \cdot \cos(\varphi_3) - O_2A \cdot \cos(\varphi_2) - 42.2 = 0 \\ O_4B \cdot \sin(\varphi_4) - AB \sin(\varphi_3) - O_2A \sin(\varphi_2) - 76.1 = 0 \end{cases}$$

* JACOBIANA

$$\Phi_q = \begin{bmatrix} O_2A \sin(\varphi_2) & -AB \sin(\varphi_3) & -O_4B \sin(\varphi_4) \\ -O_2A \cos(\varphi_2) & -AB \cos(\varphi_3) & O_4B \cos(\varphi_4) \end{bmatrix}$$

* RESOLUCIÓN MATLAB $\rightarrow \vec{q} = \{29, 32.1028, 89.8702\}^T \text{ DEG}$

$$\Phi_q = \begin{bmatrix} 23.7557 & -53.2433 & -152.9999 \\ -42.8864 & -84.6472 & 0.2136 \end{bmatrix}$$

* VELOCIDAD ANGULAR $\rightarrow$ MATLAB $\Rightarrow \dot{\vec{q}} = \{15, -7.5685, 4.19579\}^T \frac{\text{rad}}{\text{s}}$

* ACCELERACIÓN ANGULAR

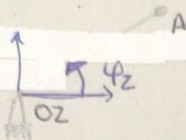
$$\ddot{\Phi}_q = \begin{bmatrix} O_2A \cos(\varphi_2) \cdot \ddot{\varphi}_2 & -AB \cos(\varphi_3) \ddot{\varphi}_3 & -O_4B \cos(\varphi_4) \ddot{\varphi}_4 \\ O_2A \sin(\varphi_2) \cdot \ddot{\varphi}_2 & AB \sin(\varphi_3) \ddot{\varphi}_3 & -O_4B \sin(\varphi_4) \ddot{\varphi}_4 \end{bmatrix}$$

$$\ddot{\Phi}_q = \begin{bmatrix} 642.8455 & 640.6524 & -1.0591 \\ 356.3351 & -402.9720 & -758.5580 \end{bmatrix}$$

* MATLAB $\rightarrow \ddot{\vec{q}} = \{10, 49.7219, 15.5481\}^T \frac{\text{rad}}{\text{s}^2}$

* BARRA 2 [RP]

(Q1)

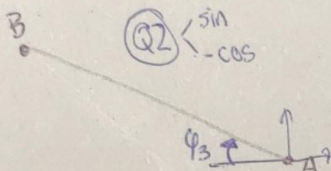


$$\vec{V}_A = \vec{\omega}_2 \times \vec{r}_{O2A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 15 \\ 49.29 & 49.29 & 0 \end{vmatrix}$$

$$[\vec{V}_A = -356,3351\hat{i} + 642,8455\hat{j} \frac{\text{mm}}{\text{s}}]$$

* BARRA 3 [MG]

(Q2)



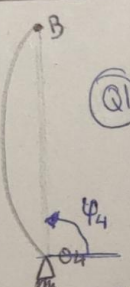
$$\vec{V}_B^{(u)} = \vec{V}_A + \vec{\omega}_3 \times \vec{r}_{AB} = \vec{V}_A + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega_3 \\ -100\cos 32,1028 & 100\sin 32,1028 & 0 \end{vmatrix} \rightarrow$$

$$\vec{V}_B^{(u)} = [-356,3351 - 53,144\omega_3]\hat{i} + [642,8455 - 84,7096\omega_3]\hat{j}$$

* BARRA 4

RP

(Q1)

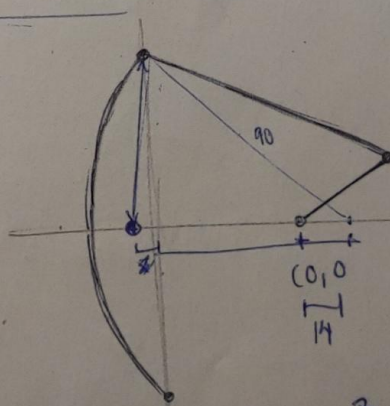
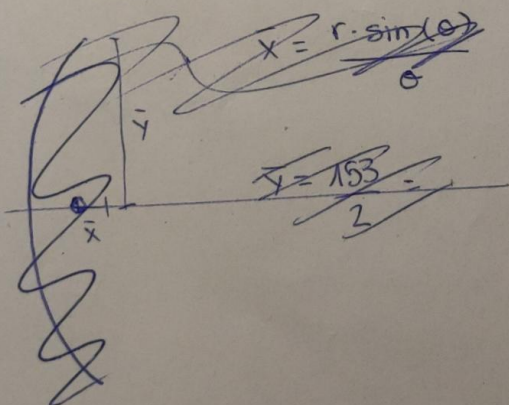


$$\vec{V}_B^{(u)} = \vec{\omega}_4 \times \vec{r}_{O4B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega_4 \\ 153\cos 32,1028 & 153\sin 32,1028 & 0 \end{vmatrix} \rightarrow$$

$$\vec{V}_B^{(u)} = -152,9996\omega_4\hat{i} + 0,3466\omega_4\hat{j}$$

IGUALDAD $\vec{V}_B = \vec{V}_B^{(u)} \rightarrow \begin{cases} \hat{i} : -356,3351 - 53,144\omega_3 = -152,9996\omega_4 \\ \hat{j} : 642,8455 - 84,7096\omega_3 = 0,3466\omega_4 \end{cases} \rightarrow$

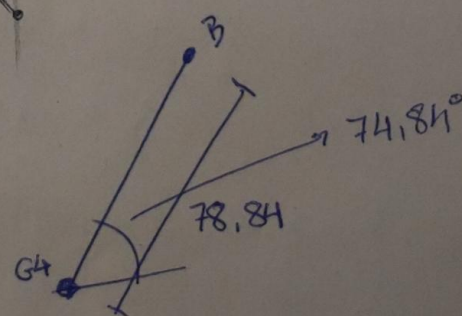
$\rightarrow \begin{cases} \omega_3 = 7,5685 \text{ rad/s} \\ \omega_4 = 4,9576 \text{ rad/s} \end{cases} \rightarrow$ Resultado sacado del modo resolver ecs. lineales para 2 incógnitas.



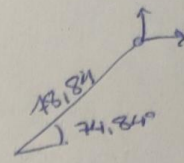
$$\bar{y} = 0$$

$$\bar{x} = 90 \cdot \frac{\sin(56)}{\frac{557}{180}} - 14$$

$$\bar{x} = 76,8$$



$$\begin{bmatrix} \vec{V}_B^{(1)} = -758,5555 \hat{i} + 1,7209 \hat{j} \frac{\text{mm}}{\text{s}} \\ \vec{V}_B^{(4)} = -758,5567 \hat{i} + 1,7184 \hat{j} \frac{\text{mm}}{\text{s}} \end{bmatrix}$$



Q3

$$\vec{V}_{GH} = \vec{\omega}_4 \times \vec{r}_{BG4} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4,9579 \\ -78,84 & 74,84 & -78,84 \end{vmatrix} \rightarrow$$

$$\rightarrow \vec{V}_{GH} = 377,2779 \hat{i} - 102,2214 \hat{j} \frac{\text{mm}}{\text{s}}$$

ACELERACIONES

* BARRA 2 Q1 y RP

$$\vec{a}_A = \vec{\alpha}_2 \times \vec{r}_{OA} + \vec{\omega}_2 \times (\vec{\omega}_2 \times \vec{r}_{OA}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 10 \\ 49C_{29} & 49S_{29} & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 15 \\ -356,3351 & 642,8455 & 0 \end{vmatrix} \rightarrow$$

$$\vec{a}_A = -593,8918 \hat{i} + 1071,4091 \hat{j} \frac{\text{mm}}{\text{s}^2}$$

* BARRA 3 Q2 y MG

$$\vec{a}_B^{(1)} = \vec{a}_A + \vec{\alpha}_3 \times \vec{r}_{AB} + \vec{\omega}_3 \times (\vec{\omega}_3 \times \vec{r}_{AB}) \rightarrow$$

$$= \vec{a}_A + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \alpha_3 \\ -100C_{32,1028} & 100S_{32,1028} & 0 \end{vmatrix} + (7,5685 \hat{k}) \times \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 7,5685 \\ -100C_{32,1028} & 100S_{32,1028} & 0 \end{vmatrix} \rightarrow$$

$$\vec{a}_B^{(1)} = [-53,144\alpha_3 + 4852,3513] \hat{i} + [-84,7096\alpha_3 - 3044,2047] \hat{j}$$

* BARRA 4 Q1 y RP

$$\vec{a}_B^{(4)} = \vec{\alpha}_4 \times \vec{r}_{OB} + \vec{\omega}_4 \times (\vec{\omega}_4 \times \vec{r}_{OB}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \alpha_4 \\ 153C_{39,8702} & 153S_{39,8702} & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4,9579 \\ -758,5567 & 1,7184 & 0 \end{vmatrix} \rightarrow$$

$$\vec{a}_B^{(4)} = [-152,9996\alpha_4 - 8,5197] \hat{i} + [0,3466\alpha_4 - 3760,8483] \hat{j}$$

$$\vec{a}_B^{(1)} = \vec{a}_B^{(4)} \rightarrow \begin{cases} \hat{i}: -53,144\alpha_3 + 4852,3513 = -152,9996\alpha_4 - 8,5197 \\ \hat{j}: -84,7096\alpha_3 - 3044,2047 = 0,3466\alpha_4 - 3760,8483 \end{cases}$$

$$\alpha_3 = 8,5778 \text{ rad/s}^2 \quad \alpha_4 = -28,791 \text{ rad/s}^2$$

$$\vec{a}_{GH} = \vec{\alpha}_4 \times \vec{r}_{OGH} + \vec{\omega}_4 \times (\vec{\omega}_4 \times \vec{r}_{OGH}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -28,791 \\ -78,84 & 74,84 & -78,84 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4,9579 \\ 377,2779 & -102,2214 & 0 \end{vmatrix} \rightarrow$$

$$\vec{a}_{GH} = -1684,0855 \hat{i} + 2464,1153 \hat{j} \frac{\text{mm}}{\text{s}^2}$$